indiv_region_increases

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This script test whether non-EUCA per-capita arrest rates rose more than EUCA per-capita rates in all states and AORs and with all denominators

```r
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(aor, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("aor", "region"))

  # aggregate regions if reuqested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(pick(-c(n_arrests, pop))) %>%
      summarize(n_arrests = sum(n_arrests),
                pop = sum(pop))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(aor, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # population map
  pop_map <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")
```

```r
  # checksum value
  checksum <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

  # checksum
  total_aggregated <- data_pop %>%
    group_by_at(geo_keys) %>%
    summarize(pop = first(pop), .groups = "drop") %>%
    pull(pop) %>%
    sum(na.rm = TRUE)
  total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
  if (!(dplyr::near(total_checksum, total_aggregated))) {
    warning("Checksum Failed!")
  }

  return(data_pop)
}
```

MPI, AOR

```r
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
```

```
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                aor diff_noneuca   diff_euca        diff
## 1    San Francisco    0.2346120  0.17335922  0.06125274
## 2      Los Angeles    0.9107710  0.78647768  0.12429332
## 3           Boston    0.1222781 -0.04000000  0.16227806
## 4          Seattle    0.1936937 -0.14705882  0.34075252
## 5          Buffalo    0.3210973 -0.03733806  0.35843538
## 6        Baltimore    1.4776536  1.00000000  0.47765363
## 7    New York City    0.9140687  0.22869211  0.68537658
## 8           Newark    2.0290828  1.06896552  0.96011726
## 9           Dallas    8.7422661  6.02696258  2.71530353
## 10       San Diego    6.3094339  2.90393661  3.40549731
## 11          Denver    5.0460829  1.36842105  3.67766190
## 12      Washington    6.0054496  1.50000000  4.50544959
## 13     San Antonio    6.1474507  1.33713376  4.81031689
## 14         Chicago    6.0683572  0.55670103  5.51165619
## 15         Phoenix    6.0034483  0.26666667  5.73678161
## 16  Salt Lake City    8.0924370  2.04000000  6.05243697
## 17         Atlanta    7.1219029  1.04918033  6.07272255
## 18    Philadelphia    7.3134921  0.56756757  6.74592450
## 19        St. Paul    7.3467337  0.42105263  6.92568104
## 20         Houston    9.3981837  2.23455240  7.16363127
## 21           Miami   13.7493495  3.59154930 10.15780023
## 22       Harlingen   15.1463813  2.41050312 12.73587821
## 23         Detroit   14.7916667  0.80769231 13.98397436
## 24     New Orleans   19.5393519  1.66666667 17.87268519
## 25         El Paso   26.7253745  2.78101102 23.94436344
```

ACS, AOR

```r
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, aor)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, aor) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##               aor diff_noneuca    diff_euca       diff
## 1   San Francisco   0.07236487  0.022717697 0.04964718
## 2          Boston   0.04465381 -0.006160518 0.05081433
## 3         Seattle   0.06600311 -0.029794182 0.09579729
## 4   New York City   0.13661890  0.021340618 0.11527828
## 5     Los Angeles   0.24110148  0.118801844 0.12229964
## 6         Buffalo   0.28047689 -0.009438593 0.28991548
```

```
## 7          Newark   0.48553382  0.098818009 0.38671581
## 8       Baltimore   0.58147907  0.157853197 0.42362587
## 9        St. Paul   1.85695596  0.075723156 1.78123281
## 10      San Diego   2.17869500  0.378582335 1.80011266
## 11     Washington   2.05910683  0.258242231 1.80086460
## 12   Philadelphia   2.07534061  0.100763403 1.97457720
## 13        Chicago   2.09238822  0.091771340 2.00061688
## 14        Detroit   2.08525869  0.075251555 2.01000713
## 15         Denver   2.30217098  0.256620311 2.04555067
## 16        Phoenix   2.16946770  0.030055979 2.13941172
## 17         Dallas   3.87335292  1.064152057 2.80920086
## 18 Salt Lake City   3.30795214  0.361899761 2.94605238
## 19        Houston   3.27111880  0.288275067 2.98284374
## 20    San Antonio   3.26119943  0.171425009 3.08977442
## 21        Atlanta   3.42829609  0.217798196 3.21049789
## 22          Miami   3.83072696  0.458699919 3.37202704
## 23      Harlingen   4.77777877  1.225677953 3.55210081
## 24        El Paso   8.95911611  0.421213862 8.53790225
## 25    New Orleans   9.13796146  0.392513973 8.74544748
```

```r
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(state, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # aggregate regions if reuqested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(state, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # checksum value
  checksum <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")
```

```r
  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("state", "region"))

  # population map
  pop_map <- data_pop %>%
    group_by(state, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

  # checksum
  total_aggregated <- data_pop %>%
    group_by_at(geo_keys) %>%
    summarize(pop = first(pop), .groups = "drop") %>%
    pull(pop) %>%
    sum(na.rm = TRUE)
  total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
  if (!(dplyr::near(total_checksum, total_aggregated))) {
    warning("Checksum Failed!")
  }

  return(data_pop)
}
```

MPI, state

```r
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv") %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```r
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                 state diff_noneuca   diff_euca        diff
## 1       MASSACHUSETTS   -0.2485549 -0.23809524 -0.01045968
## 2        RHODE ISLAND    1.0000000  1.00000000  0.00000000
## 3         CONNECTICUT    0.1306818  0.09090909  0.03977273
## 4            OKLAHOMA    0.1188119  0.00000000  0.11881188
## 5              OREGON    0.2720588  0.11764706  0.15441176
## 6          WASHINGTON    0.1144781 -0.29787234  0.41235045
## 7          CALIFORNIA    1.0938059  0.65811966  0.43568629
## 8            NEW YORK    0.7402945  0.14772727  0.59256724
## 9            MARYLAND    1.6061453  1.00000000  0.60614525
## 10           ILLINOIS    1.0575816  0.24242424  0.81515733
## 11         NEW JERSEY    2.0111857  1.06896552  0.94222017
## 12             ALASKA    1.5555556  0.50000000  1.05555556
## 13             HAWAII    1.9642857  0.64285714  1.32142857
## 14           COLORADO    2.1698113  0.55555556  1.61425577
## 15            VERMONT    2.0000000  0.00000000  2.00000000
## 16           DELAWARE    1.1071429 -1.00000000  2.10714286
## 17       PENNSYLVANIA    2.4953704  0.11764706  2.37772331
## 18          MINNESOTA    1.9560440 -0.60000000  2.55604396
## 19     NORTH CAROLINA    4.5046948  1.50000000  3.00469484
## 20          WISCONSIN    4.7000000  1.33333333  3.36666667
## 21           VIRGINIA    5.6088235  1.60000000  4.00882353
## 22             NEVADA    6.8709677  2.60000000  4.27096774
## 23           NEBRASKA    6.4313725  1.50000000  4.93137255
## 24              TEXAS   10.0156087  4.33333333  5.68227541
## 25            ARIZONA    6.0310345  0.26666667  5.76436782
## 26            GEORGIA    6.5438596  0.66666667  5.87719298
## 27              IDAHO    7.6969697  1.25000000  6.44696970
## 28      NEW HAMPSHIRE    7.8750000  0.40000000  7.47500000
```

```
## 29                 UTAH   10.1968504  1.80000000    8.39685039
## 30 DISTRICT OF COLUMBIA    9.5384615  0.75000000    8.78846154
## 31               KANSAS   11.5063291  2.00000000    9.50632911
## 32              FLORIDA   13.7701648  3.60563380   10.16453098
## 33            LOUISIANA   11.9569892  1.00000000   10.95698925
## 34                MAINE   10.0000000 -1.00000000   11.00000000
## 35              INDIANA   14.7983193  3.12500000   11.67331933
## 36             MICHIGAN   12.0634921  0.31250000   11.75099206
## 37             KENTUCKY   12.8035714  0.60000000   12.20357143
## 38                 IOWA   13.3409091  0.80000000   12.54090909
## 39                 OHIO   16.9135802  1.60000000   15.31358025
## 40       SOUTH CAROLINA   17.8425197  0.92307692   16.91944276
## 41             ARKANSAS   19.0000000  1.12500000   17.87500000
## 42            TENNESSEE   19.2603550  1.27272727   17.98762776
## 43          MISSISSIPPI   21.2608696  3.00000000   18.26086957
## 44             MISSOURI   23.0175439 -0.50000000   23.51754386
## 45           NEW MEXICO   33.0833333  2.00000000   31.08333333
## 46              ALABAMA   36.5303030  3.20000000   33.33030303
## 47              MONTANA   58.0000000  2.00000000   56.00000000
## 48         NORTH DAKOTA   59.0000000  0.00000000   59.00000000
## 49         SOUTH DAKOTA  297.0000000  7.00000000  290.00000000
## 50              WYOMING  614.0000000 16.00000000  598.00000000
## 51        WEST VIRGINIA 1268.0000000 19.00000000 1249.00000000
```

ACS, state

```
trajectory <- read_csv("../data/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")  %>%
  #filter(method == "CA" & convicted == 1)
  #filter(method == "CA" & convicted == 0)
  #filter(method == "LEA" & convicted == 1)
  filter(method == "LEA" & convicted == 0)
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```
trajectory_agg <- aggregate_data(trajectory, c("region", "state", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```
trajectory_agg_euca <- trajectory_agg %>%
  filter(region == "EUCA") %>%
```

```r
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_euca = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  select(diff_euca, state)

trajectory_agg %>%
  filter(region == "nonEUCA") %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff_noneuca = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  select(diff_noneuca, state) %>%
  merge(trajectory_agg_euca) %>%
  mutate(diff = diff_noneuca - diff_euca) %>%
  arrange(diff)
```

```
##                   state diff_noneuca   diff_euca        diff
## 1         MASSACHUSETTS  -0.08757896 -0.03928548 -0.04829349
## 2           CONNECTICUT   0.05456573  0.01496558  0.03960015
## 3              OKLAHOMA   0.05358913  0.00000000  0.05358913
## 4                HAWAII   0.25668549  0.19206146  0.06462403
## 5                OREGON   0.11358857  0.02351447  0.09007410
## 6            WASHINGTON   0.03639143 -0.05922241  0.09561384
## 7              NEW YORK   0.14797586  0.01706095  0.13091491
## 8               VERMONT   0.13803575  0.00000000  0.13803575
## 9                ALASKA   0.32599078  0.14099401  0.18499677
## 10           CALIFORNIA   0.31875618  0.08599806  0.23275811
## 11         RHODE ISLAND   0.41862412  0.11969239  0.29893173
## 12             ILLINOIS   0.37996227  0.04170294  0.33825933
## 13                MAINE   0.32831019 -0.04316298  0.37147317
## 14           NEW JERSEY   0.48125127  0.09881801  0.38243327
## 15             MARYLAND   0.63204246  0.15785320  0.47418927
## 16             DELAWARE   0.33322584 -0.17146776  0.50469361
## 17            MINNESOTA   0.41263498 -0.10430610  0.51694108
## 18         PENNSYLVANIA   0.69802106  0.02115686  0.67686421
## 19             COLORADO   0.99913770  0.10395659  0.89518111
## 20            WISCONSIN   1.14656985  0.14855254  0.99801730
## 21        NEW HAMPSHIRE   1.09802008  0.06964031  1.02837977
## 22             MICHIGAN   1.39282402  0.03090999  1.36191403
## 23             VIRGINIA   1.90949953  0.26337015  1.64612938
## 24               NEVADA   2.39171769  0.42620855  1.96550915
## 25       NORTH CAROLINA   2.38699405  0.30004251  2.08695155
## 26             NEBRASKA   2.34709869  0.25665155  2.09044713
## 27         NORTH DAKOTA   2.10782037  0.00000000  2.10782037
## 28              ARIZONA   2.17943654  0.03005598  2.14938056
## 29              GEORGIA   2.82259285  0.13880214  2.68379071
## 30                IDAHO   2.92589649  0.22962113  2.69627536
## 31                 OHIO   2.87934897  0.13639774  2.74295124
## 32 DISTRICT OF COLUMBIA   3.46001451  0.16757904  3.29243547
## 33                TEXAS   4.01025626  0.65946107  3.35079519
## 34              FLORIDA   3.83652633  0.46049874  3.37602759
## 35                 IOWA   3.69614769  0.16820858  3.52793912
## 36             KENTUCKY   4.21056341  0.10232273  4.10824068
## 37              MONTANA   4.44614795  0.16283993  4.28330802
## 38               KANSAS   4.73003913  0.42058777  4.30945136
```

```
## 39          INDIANA  4.93624369  0.52258617  4.41365752
## 40             UTAH  5.41501150  0.39247324  5.02253826
## 41        LOUISIANA  5.58903504  0.20524398  5.38379105
## 42         MISSOURI  6.09719260 -0.07309808  6.17029068
## 43      MISSISSIPPI  7.96092796  0.65645514  7.30447282
## 44         ARKANSAS  8.52153532  0.45042791  8.07110742
## 45        TENNESSEE  9.03235832  0.27200839  8.76034993
## 46     SOUTH DAKOTA 10.10754152  1.19986287  8.90767865
## 47   SOUTH CAROLINA  9.64415371  0.20479563  9.43935808
## 48       NEW MEXICO 11.21468927  0.21640338 10.99828589
## 49          ALABAMA 14.63571576  0.58597326 14.04974249
## 50          WYOMING 40.28607047  3.12317002 37.16290045
## 51    WEST VIRGINIA 55.53122537  2.47363625 53.05758912
```